

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 16-03-26
Time: 3 Hrs Max. Marks: 300

16-03-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-7_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

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PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A uniform string of length L and mass M hangs vertically from a rigid support. A transverse wave pulse is produced at the free lower end. The time taken by the pulse to reach the rigid support is:

A) $\sqrt{\frac{L}{g}}$ B) $2\sqrt{\frac{L}{g}}$ C) $\frac{1}{2}\sqrt{\frac{L}{g}}$ D) $\sqrt{\frac{2L}{g}}$

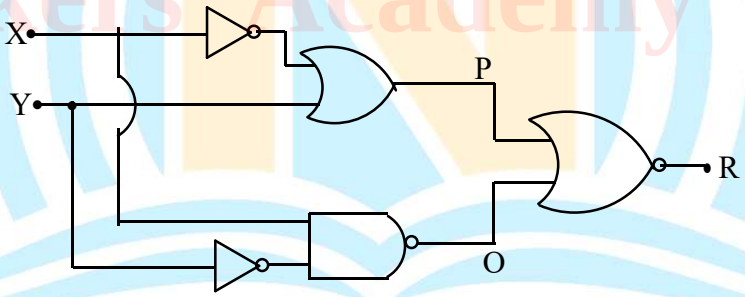
2. An electric heater of power 1000W raises the temperature of 5 kg of a liquid from 25ºC to 31ºC in 2 minutes. Heat capacity of the liquid is –

A) 2×10^4 J/ºC B) 1×10^4 J/ºC C) 3×10^4 J/ºC D) 4×10^4 J/ºC

3. If two rods of length L and 2 L having coefficients of linear expansion α and 2α respectively are connected so that total length becomes 3L, the average coefficient of linear expansion of the composite rod is

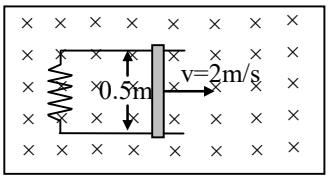
A) $\frac{3}{2}\alpha$ B) $\frac{5}{2}\alpha$ C) $\frac{5}{3}\alpha$ D) None of these

4. Figure gives a system of logic gates. From the study of truth table, it can be found that to produce a high output (1) at R, we must have –



A) X = 0, Y = 1 B) X = 1, Y = 1 C) X = 1, Y = 0 D) X = 0, Y = 0

5. A metal rod completes a circuit as shown in the figure. The circuit is normal to a magnetic field B = 0.15 T. If the resistance of the circuit is 3 ohms, then the force required to move the rod with a constant velocity of 2 m/s will be-



A) 3.75×10^{-3} N B) 3.75×10^{-2} N

C) 3.75×10^{2} N D) 3.75×10^{-4} N

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6. The frequency of incident light falling on a photo sensitive plate is doubled, then maximum kinetic energy of the emitted photoelectrons will become –

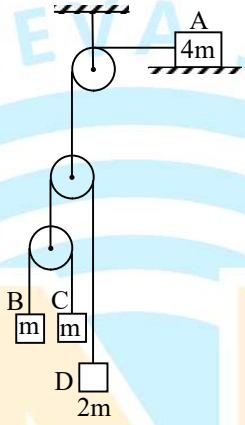
A) 2 times of the earlier value

B) More than 2 times of the earlier value

C) Less than 2 times of the earlier value

D) Unchanged

7. In the given arrangement, strings and pulleys are light and all surface are frictionless. Assuming at t = 0, system is released from rest, find the speed of block A at t = 2 sec –



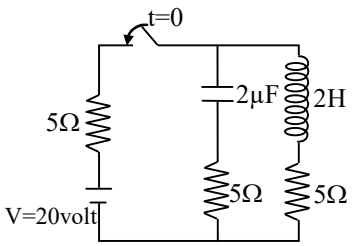
A) 8 m/s B) 9 m/s C) 10 m/s D) None of these

8. A piston shown in figure can move freely inside a non-conducting cylinder. One mole of an ideal gas $(\gamma = 1.5)$ is in left chamber and right chamber is evacuated. Initially the piston is held at middle of cylinder and temperature of gas is 300K. If the piston is released suddenly, temperature of gas will become–



A) 150 K B) $150\sqrt{2}$ K C) 300 K D) $300\sqrt{2}$ K

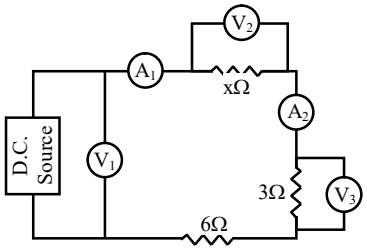
9. Calculate the ratio of current flowing through the battery at t = 0 and $t = \infty$ (t = 0 is the time of closing of switch)-



A) 1 B) 2 C) $\frac{1}{2}$ D) 0

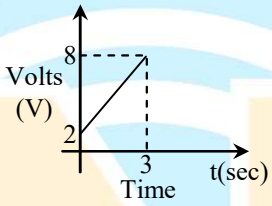
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10. In the electric circuit shown in figure, the reading of voltmeter V_1 is 26 V, and the reading of ammeter A_1 is 2 A. The value of resistance x is (Assume voltmeters and ammeters to be ideal)



A) 2Ω B) 4Ω C) 6Ω D) 8Ω

11. A circuit element is placed in a closed box. At time t = 0, a constant current generator supplying a current of 1 A is connected across the box. Potential diff. across the box varies according to graph as shown in the figure. The element in the box is -



A) a resistance of 2Ω B) a battery of emf 6V

C) an inductance of 2H D) a capacitance

12. A charge Q is distributed over two concentric hollow spheres of radii r and R (> r) such the surface densities are equal. Find the potential at the common centre

A) $\frac{Q}{4\pi\varepsilon_0}\times\frac{(r+R)}{(R+r)^2}$ B) $\frac{Q(R^2+r)^2}{4\pi\varepsilon_0(r+R)}$

C) $\frac{Q(r+R)}{4\pi\varepsilon_0(R^2+r^2)}$ D) none of these

13. A S_1 B S_2 C

In the figure shown, S_1 and S_2 represents two stationary sources of sound having equal frequency, one observer is moving from A toward C with velocity V_0 then –

A) Beats for three positions A, B and C will be heard

B) Beats will be heard in case of A and C but not in case of B

C) Beats will be not heard for A and C but will be heard for B

D) Beats will be not heard for three positions of A, B and C

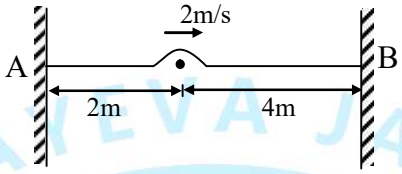
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14. A metal ball immersed in water weighs w_1 at 0°C and w_2 at 50°C. The coefficient of cubical expansion of metal is less than that of water. Then –

A) $w_1 > w_2$ B) $w_1 < w_2$

C) $w_1 = w_2$ D) data is insufficient

15. A string is tied at two rigid supports. A pulse is generated on the string as shown in figure. Minimum time after which string will regain its shape is (Neglect the time during reflection)



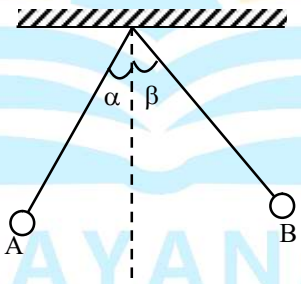
A) 2 sec B) 4 sec

C) 6 sec D) None of these

16. A liquid of mass m and specific heat C is heated to a temperature 2T. Another liquid of mass (m/2) and specific heat 2C is heated to a temperature T. If these two liquids are mixed, the resulting temperature of the mixture is -

A) (2/3)T B) (8/5)T C) (3/5)T D) (3/2)T

17. Two identical simple pendulums A and B are fixed at same point. They are displaced by very small angles α and $\beta(\beta > \alpha)$ and released from rest. Find the time after which B reaches its initial position for the first time. Collisions are elastic and length of strings is λ.



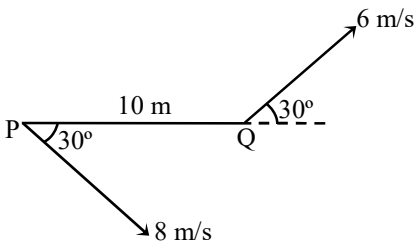
A) $\pi\sqrt{\frac{\ell}{g}}$ B) $2\pi\sqrt{\frac{\ell}{g}}$ C) $\frac{\pi\beta}{\alpha}\sqrt{\frac{\ell}{g}}$ D) $\frac{2\pi\beta}{\alpha}\sqrt{\frac{\ell}{g}}$

18. The rubber cord catapult has a cross-section area 1 mm^2 and total unstretched length 10 cm. It is stretched to 12 cm and then released to project a stone of mass 5 gm. Taking Young's modulus Y of rubber as 5×10^8 N/m^2, the velocity of projection will be -

A) 20 cm/s B) 20 m/s C) 2 m/s D) none of these

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19. Two moving particles P and Q are 10 m apart at a certain instant. The velocity of P is 8 m/s making an angle 30° with the line joining P and Q and that of Q is 6 m/s making an angle 30° with PQ as shown in the figure. Then angular velocity of P with respect to Q is



A) Zero B) 0.1 rad/sec C) 0.4 rad/sec D) 0.7 rad/sec

20. Find the de Broglie wavelength of Earth. Mass of Earth is 6×10^{24} kg. Mean orbital radius of Earth around Sun is 150×10^{6} km – (Assume Sun to be stationary)

A) 3.7 m B) 3.7×10^{-63} m C) 3.7×10^{63} m D) 3.7×10^{-63} cm

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A collimated laser beam with a diameter of 3 mm is propagating along the z-axis. To fit a specific optical sensor, the beam needs to be expanded to a final collimated diameter of 15 mm using a two-convex lens system. If the second lens in the setup has a focal length of 125 mm, what is the focal length of the first lens? (in mm)

22. Calculate the root mean square velocity of a gas of density 1.5 gram per litre at a pressure of 2×10^{6} Nm^{-2}. (Give answer in km/sec)

23. A nuclear reactor is designed to deliver 4 MW power. Nuclear fuel consists of Uranium-235 to run the reactor with amount of energy released per fission of 200 MeV. The amount of fuel needed (in grams) to run the reactor at designed rating with 100% efficiency for one year is **X**. Find the value of **X**/20 rounded off to closest integer.

24. Lifting power of helicopter depends upon hovering speed of blades (ω), length of blades (λ) and density of air (ρ). How many times lifting power will increase if hovering speed is increased to two times.

25. There are two radioactive substances A and B. Decay constant of B is two times that of A. Initially both have equal number of nuclei. After n half-lives of A, rate of disintegration of both are equal, then the value of n is

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The structure of XeO_2F_2 based on VESPR theory is best described as

A) A square planar structure with the fluorines trans to each other

B) See-saw structure with F-Xe-F bond angle close to 120^0

C) A perfect tetrahedral arrangement of substituents around Xe

D) A folded square Structure with O-Xe-O bond angle < 120^0

27. The magnetic moment of chromium in an oxide of it is $\sqrt{15}$ BM. How many electrons with azimuthal quantum number one ($l = 1$) are present in one molecule of the oxide? [Assume the oxide as an ionic compound]

A) 30 B) 42 C) 36 D) 46

28. Identify the correct order of dipole moment for the following molecules;

I. HCHO II. CH_3-CHO III. CH_3-CO-CH_3

A) I > II >III B) III > II > I C) III > I > II D) I > III > II

29. For the following two complexes, A: $[NiCl_6]^{4-}$ and B: $[NiCl_4]^{2-}$ If the ratio of CFSE of complex-A to complex-B is a/b (simplest ratio) then the value of (a + b) is?

A) 30 B) 35 C) 13 D) 4

30. The correct statement among the following that decribes the property of lanthanides is?

A) Lanthanides exhibit common oxidation states either +2 and +3 or +3 and +5

B) The compounds of lanthanides are all coloured

C) The earlier members of the series are quite reactive similar to calcium

D) They do not liberate hydrogen from dilute acids but burn in halogens to form halides.

31. For the first order reaction A(g) → 2B(g) + C(g) the half-life for decomposition of A is 3 minutes at 300K. Calculate the time (in minutes) in which partial pressure of A(g) will drop from 2 bar to 0.5 bar at 400K. Given activation energy of reaction is 840R. (ln2=0.7)

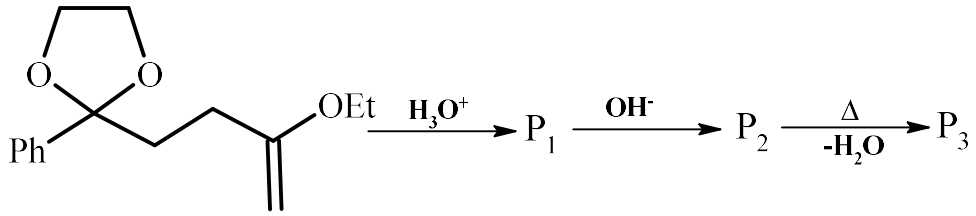
A) 1.5 B) 2 C) 3 D) 4.5

32. A solute 'X' is dissolved in a solvent with density 'd'g/cm^3. The molarity and molality of the solution are 6M and 7.5m respectively. Assuming no change in volume upon dissolution, the density 'd' of the solution is?

A) 0.75 B) 0.80 C) 1.20 D) 0.65

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33. Which of the following is correct about the major organic products P_1, P_2 and P_3 in the following reaction?



A) The product P_1 is a hydroxy ketone

B) The product P_1 gives positive iodoform test and negative Tollen's test

C) The product P_2 gives positive Lucas test and P_3 gives positive Br_2/H_2O test

D) There are six enolisable hydrogens in the product P_3

34. ΔH^0_{vap} of the process, $A_{(l)} \rightleftharpoons A_{(vap)}$ is 921.2 Calmol^{-1}. The normal boiling point of A is 100 K. When pressure is increased to 10 atm, how many times its boiling point is increased?

A) 200 B) 50 C) 10 D) 2

35. Which of the following solutions have pH equal to '5' pKa of weak acid (HA)=5

a) 100 ml, 0.1 M HA+25 ml,0.1 M NaOH

b) 100 ml, 0.1 M HA+75 ml,0.1 M NaOH+ 25ml, 0.1 M HCl

c) 100 ml, 0.1 M HA+50ml, 0.1 M HCl

d) 100 ml, 0.1 M NaA+50 ml 0.1 M HCl

A) Only a B) Only b C) 'b' and 'd' D) Only c

36. A mixture of two isomeric pentoses on reaction with hydroxylamine give 5 millimoles of a mixture of oximes. The same mixture of pentoses on treatment with bromine water produces a mixture with one of the pentoses unchanged (0.45g). The other product reacts with $NaHCO_3$. Which of the following is correct?

A) The number of millimoles of ketopentose in the original mixture is 2

B) The weight of aldopentose in the original mixture is 0.3g

C) The product formed by one of the pentose with Br_2/H_2O is aldaric acid

D) Only one of the pentoses in the mixture gives positive Tollen's test

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37. Given below are two statements:

Statement I: Action of excess Grignard reagent on carboxylic ester followed by acidification always give 3^0 alcohols.

Statement II: All aldehydes react with 2,4-DNP under optimum pH, to form a pair of isomeric condensation products.

In the light of the above statements, choose the correct answer from the options given below:

A) Both Statement I and Statement II are false

B) Statement I is false but Statement II is true

C) Both Statement I and Statement II are true

D) Statement I is true but Statement II is false

38. Match the following List-I with List-II

List-I (Compounds)		**List-II Boiling points(^{0}C)**	
(a)	Methyl acetate	(p)	213
(b)	Propionic acid	(q)	57.5
(c)	n-butyl alcohol	(r)	118
(d)	Propanamide	(s)	141

A) (a)-q, (b)-s, (c)-p, (d)-r B) (a)-r, (b)-q, (c)-s, (d)-p

C) (a)-p, (b)-s, (c)-r, (d)-q D) (a)-q, (b)-s, (c)-r, (d)-p

39. The correct order of reactivity of 2,3-dimethyl-2-butene and 2,3-dimethyl-1-butene towards reaction with bromine in presence of methanol is?

A) 2,3-dimethyl-2-butene > 2,3-dimethyl-1-butene

B) 2,3-dimethyl-2-butene < 2,3-dimethyl-1-butene

C) 2,3-dimethyl-2-butene = 2,3-dimethyl-1-butene

D) Their reactivity is not comparable

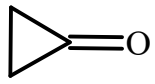
40. Compound X, $C_9H_{10}O$ is inert to Br_2/CCl_4. Vigorous oxidation with hot alkaline permanganate yields benzoic acid. X gives precipitate with semi carbazide hydrochloride and with 2,4-DNP. The number of isomeric X (including stereo isomerism) and the number of compounds among them that is/are methyl ketone, respectively are

A) 5, 1 B) 5, 0 C) 4, 1 D) 4, 0

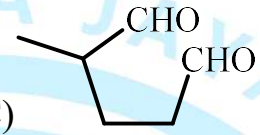
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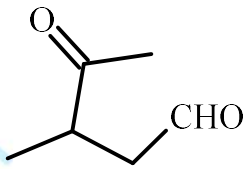
41. Compound $P(C_6H_{10})$ does not have any geometrical isomers on ozonolysis, two products $R(C_3H_4O)$ and $Q(C_3H_6O)$ are formed. 'R' gives -ve iodoform test while Q responds +vely towards $I_2 / NaOH$. 'S' is an another isomer of 'P' is an unsymmetrical hydrocarbon and on ozonolysis produces $T(C_6H_{10}O_2)$ which gives a yellow ppt with $I_2 / NaOH$ solution and also gives +ve test with Tollen's reagent. Which of the following does not represent any of the molecules amongst P, Q, R, S and T?

(Hint: P and S are cyclic compounds)

A) 

B) 

C) 

D) 

42. Identify the **correct** statements:

a) A primary standard must be non-hygroscopic and stable at room temperature.

b) NaOH is a primary standard because it can be obtained in solid pellets

c) $K_2Cr_2O_7$ and $H_2C_2O_4.2H_2O$ are examples of primary standards.

d) Secondary standards require standardization before use in precise analytical work

e) High molar mass is a desirable characteristic for a primary standard to minimize weighing errors.

Choose the **correct** options from the given options below

A) c, d and e only

B) b, c, d and e only

C) a, c, d and e only

D) a, b, c and e only

43. Given below are two statements

Statement – I: Elements 'X' and 'Y' are the most and least electronegative elements respectively among B, Al, Ga, In, Tl. The nature of the oxides X_2O_3 and Y_2O_3 is acidic and basic respectively

Statement – II: NF_3 is the most stable trihalide of Nitrogen.

In light of the above statements choose the correct answer from the options given below

A) Statement I is true but Statement II is false

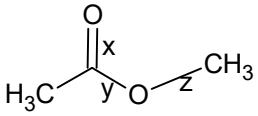
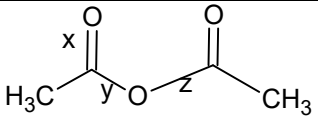
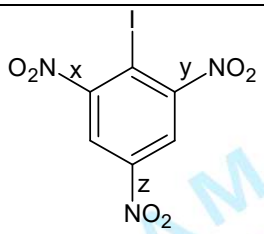
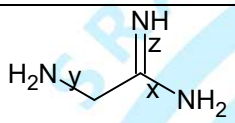
B) Both Statement I and Statement II are false

C) Statement I is false but Statement II is true

D) Both Statement I and Statement II are true

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44. Match the following compounds in **List-I** with the bond lengths in **List-II**

List-I		**List-II**	
(a)		(p)	y > x > z
(b)		(q)	x < y = z
(c)		(r)	z > y > x
(d)		(s)	z < y = x

A) (a)-r, (b)-s, (c)-q, (d)-p

B) (a)-r, (b)-q, (c)-s, (d)-p

C) (a)-p, (b)-s, (c)-r, (d)-q

D) (a)-r, (b)-s, (c)-p, (d)-q

45. Given below are two statements:

Statement I: Alcohols are dehydrated primarily in the presence of strong acid H_2SO_4 and not in the presence of HI which is almost equally acidic.

Statement II: HSO_4^- is much weaker nucleophile than I^-.

In the light of the above statements, choose the correct answer from the options given below:

A) Both Statement I and Statement II are false

B) Statement I is false but Statement II is true

C) Both Statement I and Statement II are true

D) Statement I is true but Statement II is false

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
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46. Calculate the free energy change in kJ mol^{-1} when one mole of an ionc solid MX is dissolved in water at 25°C. (Report answer to the nearest integer)

Given;

Lattice energy of MX = 780 kJ mol^{-1}

Hydration energy of MX = – 775.0 kJ mol^{-1}

Entropy of dissolution at 25°C = 45 J mol^{-1} K^{-1}

47. The mass of phenyl benzoate obtained by the Benzoylation of 47gm phenol is 'x' gm and acetanilide obtained by the acetylation of 93gm of aniline is 'y' gm. Assume both reactions are taking place with 100% yield. Find the value of $x + y =$ _____

48. The cell $Pt|H_2(1atm)|H^+(pH = x) \| I^-(1M) | AgI_{(S)}.Ag_{(S)}$ has $EMF, E_{298K} = 0$. The standard electrode potential for the reaction $AgI + e^- \rightarrow Ag + I^-$ is $-0.15V$. The value of $10x$ is _____ (Given 2.303 RT/F) = 0.06V)

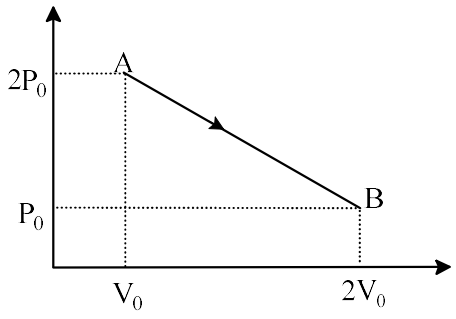
49.

20 g sample of organic compound $\xrightarrow{\text{Dumas method}}$ 2.8 L of N_2 at STP

20 g sample of organic compound $\xrightarrow{\text{Conc.}H_2SO_4}$ $(NH_4)_2SO_4$ (80%) $\xrightarrow[\text{NaOH}]{\text{excess}}$ NH_3 (100%)

If NH_3 released is completely neutralized by 200 ml 1M H_2SO_4(excess) and the remaining acid is neutralized by 100 ml of 'X' M NaOH, what is the value of X?

50. One mole of a monoatomic ideal gas is taken from point A to B as shown in figure. Change in internal energy from A to B is ____. (take P_0V_0 = 100 J]



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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. The number of points where the function

$$f(x)=\begin{cases}|2x^2-3x-7| & \text{if } x\le -1\\ [4x^2-1] & \text{if } -1<x<1\\ |x+1|+|x-2|, & \text{if } x\ge 1\end{cases}$$

$[t]$ denotes the greatest integer $\le t$, is discontinuous is ____

A) 6 B) 5 C) 7 D) 8

52. Let $S=\left[-\pi,\frac{\pi}{2}\right)-\left\{-\frac{\pi}{2},-\frac{\pi}{4},-\frac{3\pi}{4},\frac{\pi}{4}\right\}$ then the number of elements in the set

$A=\{\theta\in S:\tan\theta(1+\sqrt{5}\tan 2\theta)=\sqrt{5}-\tan 2\theta\}$ is _____

A) 5 B) 6 C) 7 D) 9

53. For $p>0$, a vector $\vec{v}_2=2\hat{i}+(p+1)\hat{j}$ is obtained by rotating the vector $\overline{v_1}=\sqrt{3}p\hat{i}+\hat{j}$ by an angle 'θ' about origin in counter clockwise direction. If $\tan\theta=\frac{\alpha\sqrt{3}-2}{4\sqrt{3}+3}$, then the values of α is equal to

A) 5 B) 6 C) 7 D) 10

54. Let m_1,m_2 be the slopes of two adjacent sides of a square of side 'a' such that $a^2+11a+3(m_1^2+m_2^2)=220$. If one vertex of the square is $(10(\cos\alpha-\sin\alpha),10(\sin\alpha+\cos\alpha))$, where $\alpha\in\left(0,\frac{\pi}{2}\right)$ and the equation of one diagonal is $(\cos\alpha-\sin\alpha)x+(\sin\alpha+\cos\alpha)y=10$,

then $72(\sin^4\alpha+\cos^4\alpha)+a^2-3a+13$ is equal to

A) 119 B) 128 C) 145 D) 155

55. Let $S=\{(x,y)\in N\times N:9(x-3)^2+16(y-4)^2\le 144\}$,

and $T=\{(x,y)\in R\times R:(x-7)^2+(y-4)^2\le 36\}$ then $n(S\cap T)$ is equal to.

A) 27 B) 36 C) 21 D) 28

56. For the system of linear equations: $2x+4y+2az=b$

$x+2y+3z=4$

$2x-5y+2z=8$

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Which of the following is **NOT correct**?

A) It has infinitely many solutions if $a=3,b=8$

B) It has infinitely many solutions if $a=3,b=6$

C) It has unique solution if $a=b=8$

D) It has unique solution if $a=b=6$

57. The number of solutions of the equation

$\log_{(x+1)}(2x^2+7x+5)+\log_{(2x+5)}(x+1)^2-4=0,(x>0)$ is ____

A) 2 B) 3 C) 1 D) 4

58. If the circles $x^2+y^2+6x+8y+16=0$ and

$x^2+y^2+2(3-\sqrt{3})x+2(4-\sqrt{6})y=k+6\sqrt{3}+8\sqrt{6},k>0$, touch internally at the point $P(\alpha,\beta)$, then $(\alpha+\sqrt{3})^2+(\beta+\sqrt{6})^2$ is equal to

A) 21 B) 20 c) 25 D) 30

59. If $\lambda x-2y=\mu$ be a tangent to the hyperbola $a^2x^2-y^2=b^2$ then

$\left(\frac{\lambda}{a}\right)^2-\left(\frac{\mu}{b}\right)^2$ is equal to _____

A) -2 B) -4 C) 2 D) 4

60. Let $\bar{a}=4\hat{i}-\hat{j}+\hat{k}$, $\bar{b}=11\hat{i}-\hat{j}+\hat{k}$ and $\bar{c}$ be a vector that $(\bar{a}+\bar{b})\times\bar{c}=\bar{c}\times(-2\bar{a}+3\bar{b})$ if $(2\bar{a}+3\bar{b})\cdot\bar{c}=1670$ then $|\bar{c}|^2$ is equal to

A) 1600 B) 1618 C) 1627 D) 1609

61. Let there be three independent events E_1,E_2 and E_3. The probability that only E_1 occurs is α, only E_2 occurs is β and only E_3 occurs is γ. Let ' p ' denote the probability of none of events occurs that satisfies the equations $(\alpha-2\beta)p=\alpha\beta$ and $(\beta-3\gamma)p=2\beta\gamma$. All the given probabilities are assumed to lie in the interval $(0,1)$.

Then, $\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3}$ is equal to _____

A) 2 B) 7 C) 5 D) 6

62. The area enclosed by the curves $y=\log_e(x+e^2)$, $x=\log_e\left(\frac{2}{y}\right)$ and y = 1 is

(A) $2+e-\log_e 2$ (B) $1+e-\log_e 2$ (C) $e-\log_e 2$ (D) $1+\log_e 2$

63. The minimum value of α for which the equation $\frac{4}{\sin x}+\frac{1}{1-\sin x}=\alpha$ has at least one solution in $\left(0,\frac{\pi}{2}\right)$ is ___ .

A) 8 B) 9 C) 3 D) 2

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64. Let $f(x)=\int \frac{dx}{(3+4x^2)\sqrt{4-3x^2}}, |x|<\frac{2}{\sqrt{3}}$. If $f(0^+)=0$ and

$f(1)=\frac{1}{\alpha\beta}\tan^{-1}\left(\frac{\alpha}{\beta}\right), \alpha,\beta>0$, then $\alpha^2+\beta^2$ is equal to ___ .

(α,β are in lowest form)

A) 28 B) 30 C) 20 D) 32

65. Let $\sum_{n=0}^{\infty}\frac{n^3((2n)!)+(2n-1)(n!)}{(n!)((2n)!)}=ae+\frac{b}{e}+c$, where $a,b,c\in\mathbb{Z}$ and $e=\sum_{n=0}^{\infty}\frac{1}{n!}$.

Then a^2-b+c is equal to ___ .

A) 27 B) 26 C) 25 D) 29

66. Let R be the set of real numbers.

Statement-1: $A=\{(x,y)\in R\times R: y-x$ is an integer $\}$ is an equivalence relation on R.

Statement-2: $B=\{(x,y)\in R\times R: x=\alpha y$ for some rational number $\alpha\}$ is an equivalence relation on R.

A) Statement-1 is true, Statement-2 is false

B) Statement-1 is false, Statement-2 is true

C) Statement-1 is true, Statement-2 is true;
Statement-2 is a correct explanation for Statement-1.

D) Statement-1 is true, Statement-2 is true;
Statement-2 is not a correct explanation for Statement-1.

67. Let z be those complex numbers which satisfy

$|z+5|\le 4$ and $z(1+i)+\bar{z}(1-i)\ge -10, i=\sqrt{-1}$.

If the maximum value of $|z+1|^2$ is $\alpha+\beta\sqrt{2}$, then the value of $(\alpha+\beta)$ is

A) 47 B) 48 C) 46 D) 45

68. The frequency distribution of the age of students in a class of 40 students is given below.

Age	15	16	17	18	19	20
No. of students	5	8	5	12	x	y

If the mean deviation about the median is 1.25, then $4x+5y$ is equal to:

A) 43 B) 44 C) 46 D) 47

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69. Let $y = f(x)$ represent a parabola with focus $\left(-\frac{1}{2}, 0\right)$ and directrix $y = -\frac{1}{2}$. Then $S = \left\{x \in R: \tan^{-1}(\sqrt{f(x)}) + \sin^{-1}(\sqrt{f(x)+1}) = \frac{\pi}{2}\right\}$

A) is an empty set
B) contains exactly one element
C) contains exactly two elements
D) is an infinite set

70. Let α and β be the roots of the quadratic equation $x^2\sin\theta - x(\sin\theta\cos\theta + 1) + \cos\theta = 0 (0° < \theta < 45°)$ and $\alpha < \beta$. Then $\sum_{n=0}^{\infty}\left(\alpha^n + \frac{(-1)^n}{\beta^n}\right)$ is equal to

A) $\frac{1}{1-\cos\theta} + \frac{1}{1+\sin\theta}$
B) $\frac{1}{1+\cos\theta} - \frac{1}{1-\sin\theta}$
C) $\frac{1}{1+\cos\theta} + \frac{1}{1-\sin\theta}$
D) $\frac{1}{1-\cos\theta} - \frac{1}{1+\sin\theta}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let $y = y(x)$ be the solution of the differential equation $(1-x^2)dy = \left(xy + (x^3+2)\sqrt{1-x^2}\right)dx, -1 < x < 1$, and $y(0) = 0$. If $\int_{\frac{-1}{2}}^{\frac{1}{2}} \sqrt{1-x^2}\,y(x)dx = k$, then k^{-1} is equal to _____

72. Let the plane P contain the line $2x + y - z - 3 = 0 = 5x - 3y + 4z + 9$ and be parallel to the line $\frac{x+2}{2} = \frac{3-y}{-4} = \frac{z-7}{5}$. Then the distance of the point $A(8,-1,-19)$ from the plane P measured parallel to the line $\frac{x}{-3} = \frac{y-5}{4} = \frac{2-z}{-12}$ is equal to ___

73. Let $f(\theta) = \sin\theta + \int_{-\pi/2}^{\pi/2}(\sin\theta + t\cos\theta)f(t)dt$. Then the value of $\left|\int_0^{\pi/2} f(\theta)d\theta\right|$ is ___.

74. If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ is equal to

75. Let $b_1b_2b_3b_4$ be a 4-element permutation with $b_i \in \{1,2,3,\ldots\ldots\ldots,100\}$ for $1 \le i \le 4$ and $b_i \neq b_j$ for $i \neq j$, such that either b_1, b_2, b_3 are consecutive integers or b_2, b_3, b_4 are consecutive integers. Then the number of such permutations $b_1b_2b_3b_4$ is equal to ___

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